

Circuit Depth and Breadth in Quantum Algorithms

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1 Introduction

In quantum computing, the resource requirements of an algorithm are commonly analyzed using two structural metrics:

- **Circuit Depth**
- **Circuit Breadth**

These metrics describe the temporal and spatial complexity of a quantum circuit.

2 Circuit Breadth (Width)

Circuit breadth, also called **circuit width**, is the number of qubits required to execute a quantum algorithm.

$$\text{Breadth} = n$$

where n is the total number of qubits (including data qubits and ancilla qubits).

Interpretation

If a quantum circuit acts on n qubits, its state lives in a Hilbert space of dimension:

$$2^n$$

Thus, breadth determines the size of the quantum state space and the hardware requirement.

Example

If an algorithm uses:

- 5 data qubits
- 2 ancilla qubits

Then,

$$\text{Breadth} = 7$$

3 Circuit Depth

Circuit depth measures the number of sequential layers of gates that must be applied.

Formally,

$$\text{Depth} = \text{Number of time steps}$$

Two gates can be placed in the same layer if they act on disjoint sets of qubits.

Parallelism Rule

If gates U_1 and U_2 act on different qubits:

$$[U_1, U_2] = 0$$

they can be executed in parallel and count as one depth layer.

Example

Consider the circuit:

1. Apply H on qubit 1
2. Apply X on qubit 2
3. Apply CNOT(1,2)

The first two gates can run in parallel, so:

$$\text{Depth} = 2$$

Layer 1: H and X

Layer 2: CNOT

4 Mathematical View

A quantum circuit implements a unitary transformation:

$$U = U_d U_{d-1} \cdots U_1$$

where each U_i represents a layer of parallel gates.

Then:

$$\text{Depth} = d$$

5 Why Depth Matters

On real quantum hardware:

- Qubits suffer from decoherence.
- Gates introduce noise.

Since errors accumulate over time, smaller depth generally improves success probability.

6 Depth vs Breadth Tradeoff

Often there is a tradeoff:

- Reducing depth may require more ancilla qubits (increase breadth).
- Reducing qubits may increase sequential operations (increase depth).

Thus, quantum algorithm design involves optimizing both.

7 Summary

- **Circuit Breadth** = Number of qubits used.
- **Circuit Depth** = Number of sequential gate layers.

Both are fundamental complexity measures of quantum algorithms.